Week-3: Propositional Equivalences

Tautologies, Logical Equivalences, De Morgan's Laws, and Introduction to Argument Forms & Validity

CS-123: Quantitative Reasoning–I

SEECS, NUST Islamabad

Tautologies, Contradictions & Contingencies

Definition 1

Definition 1

A compound proposition that is **always true**, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**.

A compound proposition that is **always false** is called a **contradiction**.

A compound proposition that is **neither** a tautology nor a contradiction is called a **contingency**.

Example: Truth Table for $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
T	F	T
F	T	T
F	T	T

Example: Truth Table for $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
T	F	T
F	T	T
F	T	T

Therefore, $p \vee \neg p$ is a TAUTOLOGY.

Example: Truth Table for $p \wedge \neg p$

p	$\neg p$	$p \wedge \neg p$
T	F	F
T	F	F
F	T	F
F	T	F

Example: Truth Table for $p \wedge \neg p$

p	$\neg p$	$p \wedge \neg p$
T	F	F
T	F	F
F	T	F
F	T	F

Therefore, $p \wedge \neg p$ is a CONTRADICTION.

Example: A Contingency

New Example

Classify $p \to q$ using a truth table.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Example: A Contingency

New Example

Classify $p \to q$ using a truth table.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Some rows T, some rows F $\Rightarrow p \to q$ is a **CONTINGENCY** – neither always true nor always false.

Pratice

✎ Try These

Using truth tables, classify each as a tautology, contradiction, or contingency.

1. $(p \land q) \to p$
2. $q \land (p \land \neg p)$
3. $p \leftrightarrow \neg p$

Pratice

✎ Try These

Using truth tables, classify each as a tautology, contradiction, or contingency.

1. $(p \wedge q) \rightarrow p$
2. $q \wedge (p \wedge \neg p)$
3. $p \leftrightarrow \neg p$

Answers: 1) Always T $\Rightarrow$ **tautology**. 2) Always F (since $p \wedge \neg p$ is always F) $\Rightarrow$ **contradiction**. 3) Always F (a statement can never have the opposite truth value of itself) $\Rightarrow$ actually this is also a **contradiction**!

Part 2

Logical Equivalences

What Are Logical Equivalences?

Informal Idea

Compound propositions that have the **same truth values in all possible cases** are called **logically equivalent**.

- Two propositions can *look* completely different but always behave exactly the same way.
- This idea is the foundation for **simplifying** logical expressions and for simplifying real code conditions and digital circuits.

Definition 2

Definition 2

The compound propositions p and q are called **logically equivalent** if $p \leftrightarrow q$ is a **tautology**.

The notation $p \equiv q$ denotes that p and q are logically equivalent.

Note

The symbol $\equiv$ is **not** a logical connective, and $p \equiv q$ is **not** a compound proposition – it is the *statement* that $p \leftrightarrow q$ is a tautology.

The symbol $\Leftrightarrow$ is sometimes used instead of $\equiv$.

The Equivalence Test

Logical Equivalence – The Test

The compound propositions p and q are equivalent **if and only if** the columns giving their truth values **agree** – row for row, for every combination of truth values.

In Practice

Build a truth table with both expressions as final columns. If their columns are **identical** in every row, the expressions are equivalent.

Part 3

De Morgan’s Laws

De Morgan's Laws

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

In Words

"NOT (both)" is the same as "NOT one, OR NOT the other."
"NOT (either)" is the same as "NOT one, AND NOT the other."

Example: Proving a De Morgan Law

Example

Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T					
T	F					
F	T					
F	F					

Example: Proving a De Morgan Law

Example

Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T				
T	F	T				
F	T	T				
F	F	F				

Solution: Building the Table (Step 1)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T				
T	F	T				
F	T	T				
F	F	F				

Solution: Building the Table (Step 1)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F			
T	F	T				
F	T	T				
F	F	F				

Solution: Building the Table (Step 1)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F			
T	F	T	F			
F	T	T				
F	F	F				

Solution: Building the Table (Step 1)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F			
T	F	T	F			
F	T	T	F			
F	F	F				

Solution: Building the Table (Step 1)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F			
T	F	T	F			
F	T	T	F			
F	F	F	T			

Solution: Building the Table (Step 2)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F			
T	F	T	F			
F	T	T	F			
F	F	F	T			

Solution: Building the Table (Step 2)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F		
T	F	T	F	F		
F	T	T	F	T		
F	F	F	T	T		

Solution: Building the Table (Step 2)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	
T	F	T	F	F	T	
F	T	T	F	T	F	
F	F	F	T	T	T	

Solution: Building the Table (Step 2)

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Conclusion: Columns Agree!

p	q	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

Conclusion

Because the truth values of $\neg(p \vee q)$ and $\neg p \wedge \neg q$ **agree** for all possible combinations of truth values of p and q, it follows that $\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$ is a **tautology**, and these compound propositions are **logically equivalent**.

Is the Biconditional a Tautology?

Verification

Show directly that $\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$ is a tautology.

$\neg(p \vee q)$	$\neg p \wedge \neg q$	$\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$
F	F	T
F	F	T
F	F	T
T	T	T

Is the Biconditional a Tautology?

Verification

Show directly that $\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$ is a tautology.

$\neg(p \vee q)$	$\neg p \wedge \neg q$	$\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q)$
F	F	T
F	F	T
F	F	T
T	T	T

Every row is T – confirmed tautology!

Practice: The Other De Morgan Law

✎ Try This

Use a truth table to show $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

Practice: The Other De Morgan Law

✎ Try This

Use a truth table to show $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
Hint: Build columns for $p, q, p \wedge q, \neg(p \wedge q), \neg p, \neg q, \neg p \vee \neg q$. You'll find $\neg(p \wedge q)$ is F only when both p, q are T, and $\neg p \vee \neg q$ is also F only when both are T – the columns will match in all four rows.

Part 4

More Equivalence Proofs

Example: The Implication Equivalence

Example

Show that $p \to q$ and $\neg p \vee q$ are logically equivalent.

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T			
T	F			
F	T			
F	F			

Example: The Implication Equivalence

Example

Show that $p \to q$ and $\neg p \vee q$ are logically equivalent.

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T		
T	F	F		
F	T	T		
F	F	T		

Solution: Filling In the Rest

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T		
T	F	F		
F	T	T		
F	F	T		

Solution: Filling In the Rest

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	
T	F	F	F	
F	T	T	T	
F	F	T	T	

Solution: Filling In the Rest

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Solution: Filling In the Rest

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The columns for $p \to q$ and $\neg p \vee q$ agree in every row – they are logically equivalent!

Why This Matters

The Implication Equivalence

$$p \to q \equiv \neg p \vee q$$

- This is one of the **most useful** equivalences in this course – it turns an “if-then” into an “or,” which is often easier to manipulate algebraically.
- We’ll use it repeatedly when **deriving** new equivalences later in this lecture.

Example: The Distributive Law

Example

Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. *(This is the distributive law of disjunction over conjunction.)*

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

Example: The Distributive Law

Example

Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent. *(This is the distributive law of disjunction over conjunction.)*

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T				
T	T	F	F				
T	F	T	F				
T	F	F	F				
F	T	T	T				
F	T	F	F				
F	F	T	F				
F	F	F	F				

Solution: The Full 8-Row Table

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T				
T	T	F	F				
T	F	T	F				
T	F	F	F				
F	T	T	T				
F	T	F	F				
F	F	T	F				
F	F	F	F				

Solution: The Full 8-Row Table

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T			
T	T	F	F	T			
T	F	T	F	T			
T	F	F	F	T			
F	T	T	T	T			
F	T	F	F	F			
F	F	T	F	F			
F	F	F	F	F			

Solution: The Full 8-Row Table

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	
T	T	F	F	T	T	T	
T	F	T	F	T	T	T	
T	F	F	F	T	T	T	
F	T	T	T	T	T	T	
F	T	F	F	F	T	F	
F	F	T	F	F	F	T	
F	F	F	F	F	F	F	

Solution: The Full 8-Row Table

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Solution: The Full 8-Row Table

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Columns 5 and 8 agree in every row – logically equivalent!

Practice: Verify Two More Laws

✎ Try These

Use truth tables to verify:

1. **Commutative Law:** $p \vee q \equiv q \vee p$
2. **Idempotent Law:** $p \vee p \equiv p$

Part 5

Using De Morgan’s Laws in English

Recap: De Morgan's Laws

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Why Learn This in English?

De Morgan's Laws are exactly how we correctly negate compound statements like "*A* and *B*" or "*A* or *B*" in everyday language – a skill you'll use constantly when writing careful code conditions.

Example: Negating “Miguel has a cellphone AND a laptop”

Example

Use De Morgan’s Laws to express the negation of: “Miguel has a cellphone and he has a laptop computer.”

Example: Negating “Miguel has a cellphone AND a laptop”

Example

Use De Morgan’s Laws to express the negation of: “Miguel has a cellphone and he has a laptop computer.”

Let p: “Miguel has a cellphone,” q: “Miguel has a laptop computer.” The statement is $p \wedge q$.

Example: Negating “Miguel has a cellphone AND a laptop”

Example

Use De Morgan’s Laws to express the negation of: “Miguel has a cellphone and he has a laptop computer.”

Let p: “Miguel has a cellphone,” q: “Miguel has a laptop computer.” The statement is $p \wedge q$.

$$\neg(p \wedge q)$$

Example: Negating “Miguel has a cellphone AND a laptop”

Example

Use De Morgan’s Laws to express the negation of: “Miguel has a cellphone and he has a laptop computer.”

Let p: “Miguel has a cellphone,” q: “Miguel has a laptop computer.” The statement is $p \wedge q$.

$$\neg(p \wedge q)$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad \text{(De Morgan’s Laws)}$$

Solution: The Negation in English

✓ Solved

"Miguel has a cellphone and he has a laptop computer."

$\neg(p \wedge q) \equiv \neg p \vee \neg q$ (De Morgan's Laws)

The negation is therefore:

"Miguel does not have a cellphone *or* he does not have a laptop computer."

Example: Negating “Heather OR Steve will go”

Example

Use De Morgan’s Laws to express the negation of: “Heather will go to the concert or Steve will go to the concert.”

Example: Negating “Heather OR Steve will go”

Example

Use De Morgan’s Laws to express the negation of: “Heather will go to the concert or Steve will go to the concert.”

Let p: “Heather will go to the concert,” q: “Steve will go to the concert.” The statement is $p \vee q$.

Example: Negating “Heather OR Steve will go”

Example

Use De Morgan’s Laws to express the negation of: “Heather will go to the concert or Steve will go to the concert.”

Let p: “Heather will go to the concert,” q: “Steve will go to the concert.” The statement is $p \vee q$.

$$\neg(p \vee q)$$

Example: Negating “Heather OR Steve will go”

Example

Use De Morgan’s Laws to express the negation of: “Heather will go to the concert or Steve will go to the concert.”

Let p: “Heather will go to the concert,” q: “Steve will go to the concert.” The statement is $p \vee q$.

$$\neg(p \vee q)$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad \text{(De Morgan’s Laws)}$$

Solution: The Negation in English

✓ Solved

"Heather will go to the concert or Steve will go to the concert."

$\neg(p \vee q) \equiv \neg p \wedge \neg q$ (De Morgan's Laws)

The negation is therefore:

"Heather will not go to the concert *and* Steve will not go to the concert."

Practice: Negate These Statements

✎ Unsolved – Try These

Use De Morgan's Laws to find the negation of each.

1. "Jan is rich and happy."
2. "Carlos will bicycle or run tomorrow."

Practice: Negate These Statements

✎ Unsolved – Try These

Use De Morgan's Laws to find the negation of each.

1. "Jan is rich and happy."
2. "Carlos will bicycle or run tomorrow."

Answers: 1) "Jan is not rich or Jan is not happy."
2) "Carlos will not bicycle and Carlos will not run tomorrow."

Practice: Two More Negations

✎ Unsolved – Try These

1. “Mei walks or takes the bus to class.”
2. “Ibrahim is smart and hard working.”

Common Mistake

Don’t forget to **flip the connective too**: negating AND gives OR, and negating OR gives AND – not just negating each piece!

Practice: Two More Negations

✎ Unsolved – Try These

1. “Mei walks or takes the bus to class.”
2. “Ibrahim is smart and hard working.”

Answers: 1) “Mei does not walk and does not take the bus to class.”
2) “Ibrahim is not smart or not hard working.”

Common Mistake

Don’t forget to **flip the connective too**: negating AND gives OR, and negating OR gives AND – not just negating each piece!

Part 6

Constructing New Logical Equivalences

Why Build a Table of Equivalences?

- We've now **proved** several equivalences directly with truth tables.
- But truth tables get unwieldy fast – 4 variables need 16 rows, 5 variables need 32!
- Instead, mathematicians have compiled a standard **toolkit** of equivalences we can use like algebra rules – **no truth table required** once you know them.

Logical Equivalences (Part 1 of 3)

Equivalence	**Name**
$p \wedge T \equiv p \quad p \vee F \equiv p$	Identity Laws
$p \vee T \equiv T \quad p \wedge F \equiv F$	Domination Laws
$p \vee p \equiv p \quad p \wedge p \equiv p$	Idempotent Laws
$\neg(\neg p) \equiv p$	Double Negation Law
$p \vee q \equiv q \vee p \quad p \wedge q \equiv q \wedge p$	Commutative Laws

Logical Equivalences (Part 2 of 3)

Equivalence	**Name**
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative Laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive Laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's Laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption Laws

Logical Equivalences (Part 3 of 3)

Equivalence	Name
$p \vee \neg p \equiv T$	Tautology
$p \wedge \neg p \equiv F$	Contradiction (Negation Law)
$p \to q \equiv \neg p \vee q$	Implication Equivalence
$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$	Biconditional Equivalence

These Are Your Toolkit

Every equivalence proof from here on uses **only** these identities – no truth tables needed!

Quick Reference Card

Identity $p \wedge T \equiv p,\ p \vee F \equiv p$	**Domination** $p \vee T \equiv T,\ p \wedge F \equiv F$	**Idempotent** $p \vee p \equiv p,\ p \wedge p \equiv p$
Double Negation $\neg\neg p \equiv p$	**Commutative** $p \vee q \equiv q \vee p$	**Associative** $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	**De Morgan's** $\neg(p \wedge q) \equiv \neg p \vee \neg q$	**Absorption** $p \vee (p \wedge q) \equiv p$

Keep this table handy you'll use it in every derivation from here on.

Part 7

Deriving New Equivalences

The Algebraic Method

Strategy

Starting from one expression, apply **known** equivalences one step at a time – like solving an algebra equation – until you reach the target expression. Each step must cite **which** law justifies it.

- This method is often **faster** than truth tables for expressions with many variables.
- It also **explains why** the equivalence holds, not just *that* it holds.

Example: Deriving $\neg(p \to q) \equiv p \wedge \neg q$

Example

Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent – **without** a truth table, using identities we already know.

$$\neg(p \to q) \equiv \neg(\neg p \vee q)$$

Example: Deriving $\neg(p \to q) \equiv p \wedge \neg q$

Example

Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent – **without** a truth table, using identities we already know.

$$\neg(p \to q) \equiv \neg(\neg p \vee q) \quad \textit{by the Implication Equivalence}$$

Step-by-Step Derivation

$$\neg(p \to q) \quad \equiv \neg(\neg p \vee q) \qquad \text{by the Implication Equivalence}$$

Step-by-Step Derivation

$$
\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{by the Implication Equivalence} \\
&\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan Law}
\end{aligned}
$$

Step-by-Step Derivation

$$
\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{by the Implication Equivalence} \\
&\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan Law} \\
&\equiv p \wedge \neg q && \text{by the Double Negation Law}
\end{aligned}
$$

Step-by-Step Derivation

$$
\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{by the Implication Equivalence} \\
&\equiv \neg(\neg p) \wedge \neg q && \text{by the second De Morgan Law} \\
&\equiv p \wedge \neg q && \text{by the Double Negation Law}
\end{aligned}
$$

$$\therefore \neg(p \to q) \equiv p \wedge \neg q$$

Why This Result Matters

A Very Useful Identity

$$\neg(p \to q) \equiv p \wedge \neg q$$

- This tells us **exactly** how to negate an if-then statement: "NOT (if p then q)" means "p happens but q doesn't."
- Example: The negation of "If it rains, I bring an umbrella" is **"It rains and I don't bring an umbrella"** – not "if it doesn't rain..."

Example:

Example

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$$

Example:

Example

Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

$$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q) \quad \textit{by the second De Morgan Law}$$

Derivation:

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the 2nd De Morgan Law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] \\
&\equiv \neg p \wedge (p \vee \neg q)
\end{aligned}
$$

Derivation:

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the 2nd De Morgan Law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the 1st De Morgan Law} \\
&\equiv \neg p \wedge (p \vee \neg q)
\end{aligned}
$$

Derivation:

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the 2nd De Morgan Law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the 1st De Morgan Law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{by the Double Negation Law}
\end{aligned}
$$

Derivation

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{by the 2nd De Morgan Law} \\
&\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && \text{by the 1st De Morgan Law} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{by the Double Negation Law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \\
&\equiv F \vee (\neg p \wedge \neg q)
\end{aligned}
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&\equiv \neg p \wedge (p \vee \neg q) && \text{by the Double Negation Law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the 2nd Distributive Law} \\
&\equiv F \vee (\neg p \wedge \neg q)
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&\equiv \neg p \wedge (p \vee \neg q) && \text{by the Double Negation Law} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{by the 2nd Distributive Law} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{by Contradiction } (\neg p \wedge p \equiv F)
\end{aligned}
$$

Derivation:

$\neg(p \vee (\neg p \wedge q))$	$\equiv \neg p \wedge \neg(\neg p \wedge q)$	by the 2nd De Morgan Law
	$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$	by the 1st De Morgan Law
	$\equiv \neg p \wedge (p \vee \neg q)$	by the Double Negation Law
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the 2nd Distributive Law
	$\equiv F \vee (\neg p \wedge \neg q)$	by Contradiction
	$\equiv (\neg p \wedge \neg q) \vee F$	
	$\equiv \neg p \wedge \neg q$	

Derivation:

$\neg(p \vee (\neg p \wedge q))$	$\equiv \neg p \wedge \neg(\neg p \wedge q)$	by the 2nd De Morgan Law
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	$\equiv \neg p \wedge (p \vee \neg q)$	by the Double Negation Law
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the 2nd Distributive Law
	$\equiv F \vee (\neg p \wedge \neg q)$	by Contradiction
	$\equiv (\neg p \wedge \neg q) \vee F$	by the Commutative Law (for $\vee$)
	$\equiv \neg p \wedge \neg q$	

Derivation:

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	$\equiv \neg p \wedge (p \vee \neg q)$	by the Double Negation Law
	$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the 2nd Distributive Law
	$\equiv F \vee (\neg p \wedge \neg q)$	by Contradiction
	$\equiv (\neg p \wedge \neg q) \vee F$	by the Commutative Law (for $\vee$)
	$\equiv \neg p \wedge \neg q$	by the Identity Law (for F)

Derivation:

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Hence, the two expressions are logically equivalent!

Practice:

✎ Try This

Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences (not a truth table).

Practice:

✎ Try This

Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences (not a truth table).

Solution sketch:

$$
\begin{aligned}
(p \wedge \neg q) \vee q &\equiv (p \vee q) \wedge (\neg q \vee q) && \text{Distributive Law} \\
&\equiv (p \vee q) \wedge T && \text{Negation Law} \\
&\equiv p \vee q && \text{Identity Law}
\end{aligned}
$$

Exercise

Exercise

Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences.

$$(p \wedge \neg q) \vee q \equiv (p \vee q) \wedge (\neg q \vee q)$$

Exercise

Exercise

Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences.

$(p \wedge \neg q) \vee q \equiv (p \vee q) \wedge (\neg q \vee q)$ Distributive Law

Exercise

Exercise

Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences.

$$
\begin{aligned}
(p \wedge \neg q) \vee q &\equiv (p \vee q) \wedge (\neg q \vee q) && \text{Distributive Law} \\
&\equiv (p \vee q) \wedge T && \text{Negation Law } (\neg q \vee q \equiv T)
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Exercise

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Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences.

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&\equiv p \vee q && \text{Identity Law}
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Prove: $(p \wedge \neg q) \vee q \equiv p \vee q$ using logical equivalences.

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&\equiv (p \vee q) \wedge T && \text{Negation Law } (\neg q \vee q \equiv T) \\
&\equiv p \vee q && \text{Identity Law}
\end{aligned}
$$

QED!

Part 9

Introduction to Argument Forms

What Is an Argument?

Definition

An **argument** is a sequence of propositions. All but the final proposition are called **premises**, and the final proposition is called the **conclusion**.

➤ Example

Premise 1: If it rains, the ground gets wet.
Premise 2: It is raining.
Conclusion: Therefore, the ground gets wet.

Notation

We often write premises above a horizontal line and the conclusion below it, with $\therefore$ (“therefore”) marking the conclusion.

What Is an Argument Form?

Definition

An **argument form** is an argument written using **propositional variables** instead of specific statements – it shows the **structure** of the reasoning, stripped of its content.

➤ The Same Argument, in Form

$p \to q$
p
———————
$\therefore q$

Why Strip Away Content?

This lets us check whether the **reasoning pattern** is trustworthy – separately from whether the specific facts are true.

From English to Argument Form

Example

"If Ali studies, he passes the exam. Ali studies. Therefore, Ali passes the exam."

- Let p: "Ali studies," q: "Ali passes the exam."
- Premise 1: $p \to q$
- Premise 2: p
- Conclusion: $\therefore q$

Same Skeleton, Different Story

Any argument of the exact form $p \to q,\ p \ \therefore q$ shares this same underlying structure – regardless of what p and q actually mean.

What Does “Valid” Mean?

Definition: Valid Argument Form

An argument form is **valid** if, whenever **all** its premises are true, the conclusion is **guaranteed** to be true too.

Careful – A Common Misunderstanding

Validity is about the **form**, not the actual truth of the premises! An argument can be **valid** even if a premise is factually false – validity only asks: *if* the premises *were* true, would the conclusion have to be true too?

Valid vs. Sound: An Important Distinction

Valid	**Sound**
The **form** guarantees: IF premises are true, THEN the conclusion must be true.	The argument is valid **AND** all its premises are **actually** true.

Example: Valid but NOT Sound

"If the moon is made of cheese, then it is edible. The moon is made of cheese. Therefore, the moon is edible." – **Valid** form ($p \rightarrow q, p \therefore q$), but **unsound** (premise 2 is false in reality).

Practice: Identify the Parts

✎ Unsolved – Try This

"All students who pass Quiz 1 study hard. Fatima passed Quiz 1. Therefore, Fatima studies hard."
Identify the premises, the conclusion, and write the argument form.

Practice: Identify the Parts

✎ Unsolved – Try This

"All students who pass Quiz 1 study hard. Fatima passed Quiz 1. Therefore, Fatima studies hard."

Identify the premises, the conclusion, and write the argument form.

Premises: (1) Students who pass Quiz 1 study hard. (2) Fatima passed Quiz 1.

Conclusion: Fatima studies hard.

Form: Let p: "you pass Quiz 1," q: "you study hard." Form: $p \to q,\ p\ \therefore q$.

Part 10

Testing Validity with Truth Tables

The Method

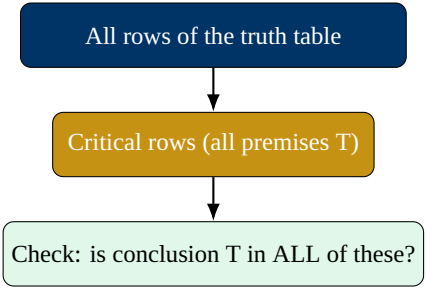
How to Test Validity

1. Build a truth table with one column for **each premise** and one for the **conclusion**.
2. Find every **“critical row”** – a row where **all** premises are **T**.
3. Check the conclusion’s value in **every** critical row.

Decision Rule

Valid: the conclusion is T in *every* critical row.
Invalid: *some* critical row has the conclusion F.

Why Does This Work?



- Rows where a premise is **false** don't matter – "if premises are true" is automatically satisfied (vacuously) when a premise isn't even true.
- We only care about the scenarios where the argument's **starting assumptions actually hold**.

A Worked Mini-Example

Testing $p \to q,\ p\ \therefore q$

Is this argument form valid?

p	q	$p \to q$	p	q (conclusion)
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

A Worked Mini-Example

Testing $p \to q,\ p\ \therefore q$

Is this argument form valid?

p	q	$p \to q$	p	q (conclusion)
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

Only Row 1 has *both* premises T (critical row). In that row, the conclusion q is also T. $\Rightarrow$ VALID.

Quick Check

✎ Try This

An argument has premises $p \vee q$ and $\neg p$, with conclusion q. There are 4 rows total. In which rows are **both** premises true?

Quick Check

✎ Try This

An argument has premises $p \vee q$ and $\neg p$, with conclusion q. There are 4 rows total. In which rows are **both** premises true?

Answer: Need $p \vee q = \text{T}$ **and** $\neg p = \text{T}$ (so $p = \text{F}$). This forces $p = \text{F}$, and $p \vee q = \text{T}$ requires $q = \text{T}$. So the **only** critical row is $p = \text{F}$, $q = \text{T}$. Check the conclusion q in that row: T. **Valid!** (This is the *Disjunctive Syllogism*, coming up next.)